

FUNCTIONAL REQUIREMENTS DOCUMENT (FRD)

Project Title:

Supply Chain Performance & Risk Analysis Dashboard for DataCo Global

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1. Project Title

Supply Chain Performance & Risk Analysis Dashboard for DataCo Global

2. Dashboard Sections

The dashboard consists of five main sections, each addressing a specific analytical goal:

- 1 Executive Overview** – Displays key KPIs and overall supply chain health at a glance.
- 2 Delivery Performance Analysis** – Focuses on On-Time Delivery rate, delays, and trends.
- 3 Profitability & Cost Analysis** – Analyzes profit margins and the impact of delays and shipping costs.
- 4 Logistics & Shipping Analysis** – Evaluates performance by shipping mode and region.
- 5 Regional & Category Insights** – Provides deep dive into regional and product category performance with actionable insights.

3. Data Requirements

Executive Overview

- Fact Table: Sales_Shipments_Fact
- Order Date, Year, Quarter, Month Name
- Delivery Status, Late_delivery_risk
- Benefit per order (Profit), Sales per customer

Delivery Performance Analysis

- Days for shipping (real)
- Days for shipment (scheduled)
- Delay Days (calculated)
- On Time Flag (calculated)
- Shipping Mode, Order Region

Profitability & Cost Analysis

- Benefit per order (Profit)
- Sales per customer
- Product Category, Order Region
- Delay Days

Logistics & Shipping Analysis

- Shipping Mode

- Order Region, Order Country
- Delay Days, Delivery Status

Regional & Category Insights

- Product Category, Department Name
- Order Region, Customer Segment
- Late_delivery_risk

4. Filters / Slicers

- Date Range (Order Date)
- Year / Quarter / Month
- Shipping Mode
- Product Category / Department Name
- Order Region / Order Country
- Customer Segment
- Delivery Status (On Time / Late)

5. Visuals / Charts

Executive Overview

- KPI Cards (OTD %, Total Profit, Total Sales, Avg Delay Days)
- Line Chart (Monthly trend of OTD % and Profit)
- Donut Chart (On Time vs Late Delivery)
- Card visuals with conditional formatting

Delivery Performance Analysis

- Stacked Bar Chart (Late deliveries by Shipping Mode)
- Line & Column Chart (Delay trend over time)
- Matrix (OTD % by Region and Category)
- Decomposition Tree (Root cause of delays)

Profitability & Cost Analysis

- Column Chart (Profit by Category and Region)
- Scatter Chart (Delay Days vs Profit)
- Waterfall Chart (Profit impact analysis)

Logistics & Shipping Analysis

- Bar Chart (Shipping Mode performance)
- Treemap or Map (by Region)
- Slicer + Card visuals

Regional & Category Insights

- Bar Chart (Top/Bottom performing regions)
- Matrix with conditional formatting
- Pie Chart (Category contribution)

6. Interactivity

- Cross-filtering enabled across all pages
- Drill-down (Year → Quarter → Month)
- Drill-through pages (click on region or category to see details)

- Bookmark navigation between sections
- Tooltips with additional context

7. Calculations / Measures

Key DAX measures to be created:

- Total Sales = SUM(Sales per customer)
- Total Profit = SUM(Benefit per order)
- Profit Margin % = DIVIDE([Total Profit], [Total Sales])
- On-Time Delivery (OTD) %
- Late Delivery Rate %
- Average Delay Days
- YoY Growth in Profit and OTD %
- Rolling 3-Month Average OTD %

8. Export / Reporting Needs

- PDF Export of dashboard views
- Excel export of detailed tables
- Shareable link via Power BI Service

9. Notes / Special Instructions

- Dashboard will follow a clean, professional color theme (Blues and Greens)
- All visuals will be optimized for performance with ~180,000 rows
- Focus on actionable business insights and recommendations
- “Benefit per order” will be treated as Profit